

Sai Sharan Mankala

AI/ML Engineer

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SUMMARY

AI/ML Engineer with 4+ years of experience building production search, retrieval, ranking, and real-time ML systems. Skilled in developing hybrid search platforms using Python, PyTorch, Elasticsearch, FAISS, BM25, dense embeddings, cross-encoder reranking, Kafka, Kubernetes, and AWS. Proven track record of improving search accuracy by 32%, reducing p95 search latency by 27%, and operating distributed search systems supporting 1M+ daily users.

PROFESSIONAL EXPERIENCE

PERPLEXITY Mar 2025 – Present

AI/ML Engineer San Francisco, CA

Built and deployed end-to-end AI search and agentic workflows—from query understanding and retrieval through reasoning, evaluation, backend services, and production deployment.

- Built and owned end-to-end AI search workflows combining transformer-based query understanding, semantic classification, routing, retrieval, and grounded response generation using Python, PyTorch, Hugging Face Transformers, and LoRA, improving response accuracy by **32%** across production systems serving **100,000+ daily users**.
- Caught a production regression where complex queries were misrouted to a lightweight path, producing shallow answers; built an automated RAGAS-based offline evaluation gate into the deployment pipeline using failed production and synthetic queries, preventing recurrence and reducing hallucination rates by 19%. • Diagnosed intermittent latency spikes in the hybrid retrieval pipeline (BM25 + dense + cross-encoder reranking), tracing the root cause to both cache invalidation issues in Redis and misrouted queries taking the slower reasoning path; fixed cache invalidation logic, added fallback handling for cache failures, and reduced p95 search latency by 27%.
- Translated complex user requests into LangGraph-based agentic workflows for multi-step retrieval, tool calling, and structured reasoning; implemented retries, checkpointing, state management, and partial-failure recovery to deliver reliable production outcomes despite incomplete or failed dependencies.
- Traffic spikes on our EKS-based inference services were causing dropped requests with no alerting to catch capacity issues before users were affected; added autoscaling policies and Prometheus/Grafana-based alerts to catch load issues early, improving request throughput while maintaining stable inference latency during peak traffic.
- Fine-tuned pretrained transformer models using LoRA through Hugging Face Transformers and PEFT to adapt query-classification models for limited labeled data, improving classification quality without the cost and time of full fine-tuning.
- Engineered real-time ingestion and indexing pipelines using Kafka, PySpark, Airflow, and AWS S3 to process continuously arriving documents, maintain retrieval-index freshness, and scale across billions of searchable documents.

AMAZON Apr 2021 – June 2024

Machine Learning Engineer India

Built and deployed large-scale demand forecasting systems powering automated inventory planning across Amazon's fulfillment network.

- Partnered with engineering and supply-chain stakeholders to translate inventory-planning requirements into probabilistic forecasting systems using PyTorch, DeepAR, and GluonTS, generating P50/P70/P90 forecasts across millions of SKUs for **3,000+ internal users**; improved forecast accuracy by **34%** and reduced stockouts by **27%** through feature engineering, backtesting, and error analysis.
- Improved forecast accuracy by 18% and reduced stockout rates by 16% by incorporating historical demand, seasonality, promotions, and regional purchasing patterns into time-series forecasting models and validating model and feature changes before production deployment.
- Developed distributed feature-engineering and data-processing pipelines using Python, Apache Spark, SQL, and Airflow to handle high-volume transactional data, late-arriving records, and inconsistent regional inputs for daily forecasting workflows.
- Automated model training, validation, and batch-inference workflows using AWS SageMaker, EC2, S3, and EMR, reducing model retraining time by 22% and enabling reproducible forecasting updates across production environments.
- Built backend prediction services using Java, Scala, REST APIs, and gRPC to deliver forecasting outputs and inventory recommendations to downstream supply-chain systems with reliable, low-latency request processing.
- Built event-driven data pipelines using Apache Kafka, AWS Lambda, DynamoDB, and Redis to process inventory updates and demand events, enabling timely synchronization across forecasting and replenishment services.

- Deployed forecasting applications using Docker, Kubernetes, CI/CD pipelines, and Amazon CloudWatch, reducing deployment time by 31% while improving production monitoring, application reliability, and operational support.
- Partnered with data scientists, software engineers, and supply-chain stakeholders to translate inventory-planning requirements into production machine-learning solutions, reducing excess inventory by 14% and improving replenishment planning by 17%.

TECHNICAL SKILLS

Programming Languages: Python, Java, Node.js, SQL, C++

Generative AI & Retrieval: RAG, LangGraph, LangChain, LlamaIndex, AI Agents, Tool Calling, Structured Outputs, RAGAS, Semantic Search, Hybrid Retrieval (BM25 + Dense), Cross-Encoder Reranking, Elasticsearch, FAISS, Information Retrieval, Search Relevance, Inverted Indexes, Offline Evaluation

Machine Learning: PyTorch, Scikit-learn, XGBoost, Transformer Models, Hugging Face, PEFT, LLMs, Fine-Tuning (LoRA, RLHF), INT8 Quantization, Time-Series Forecasting (DeepAR, GluonTS), Feature Engineering, Model Evaluation

Cloud: AWS, Docker, Kubernetes, CI/CD, FastAPI, REST APIs, gRPC, Kafka, Spark, Airflow, PostgreSQL, Redis, Prometheus, Grafana, CloudWatch

Backend & Data Engineering: FastAPI, REST APIs, gRPC, Microservices, Apache Spark, Kafka, Airflow, Redis, PostgreSQL, DynamoDB

ML Infrastructure & Model Serving: AWS SageMaker, MLflow, Weights & Biases, vLLM, Triton Inference Server, DeepSpeed, CUDA

EDUCATION

Master of Science in Computer Science

Southern Illinois University Edwardsville

Master's Project — Large-Scale AI Traffic Risk Forecasting and Hotspot Prediction Platform

- Engineered an end-to-end machine learning platform using Python, XGBoost, FastAPI, Pandas, GeoPandas, Scikit-learn, Tableau, and weather-enriched transportation datasets spanning 20+ years of California crash history to perform statewide traffic risk classification, hotspot prediction, and temporal forecasting.
- Developed scalable ETL, feature engineering, geospatial aggregation, and model training pipelines leveraging temporal features, weather signals, road characteristics, and explainable AI techniques for predictive risk modeling across California regions.
- Built AI-driven risk intelligence workflows incorporating SHAP explainability, anomaly detection, hotspot stability analysis, and predictive analytics to identify emerging high-risk zones and support proactive resource allocation.

Bachelor of Technology in Data Science

Malla Reddy University